

Computational Modelling of Somatosensory Laminar Circuits

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Abstract

Changed stuff

1 Introduction

Briefly introduce the biological background and motivation for the modelling project.

2 Methods

2.1 Model Description

2.2 Model Parameters and Units

Connectivity Matrix PSPs (Synaptic Strengths) mV W matrix : mV potential rate
Hz timestep sec tau (membrane time constant) sec

SI Units: V Hz sec

2.3 Connectivity Matrix

The connectivity value that weights the connection between two population depends on three factors: Connection probability: what is the probability that a neuron from population A connects to a neuron in population B. The probability is computed by conducting pairwise bernoulli trials. Example: we look at layer 4 and pick one PV cell and check if it connects to each of the SOM cells. Multapses (multiple synapses between the same pair of neurons) are lumped together. Synaptic strength: what is the post-synaptic potential (PSP) change induced by a single pre-synaptic spike. The synaptic strength also reflects increased potential due to multapses (since a stronger PSP would arise). Cell count: how many neurons are in population A. The thalamus cell counts are computed using the summarized cell count of one column (Area 1) The paper by [?] provides synaptic strength values, which are given as post-synaptic potential changes in mV (see Table 3 on page 6 of the preprint).

The connectivity value is then calculated as:

$$\text{Connectivity Value} = \text{Synaptic Strength} \times \text{Connection Probability} \times \text{Cell Count} \quad (1)$$

Table 1 summarizes the sources of the values used for each factor that build the connectivity matrix.

Table 1: Sources of values for the connectivity matrix.

	Synaptic strength	Connection probability	Cell count
Area 1	Jiang et al. 2023	Jiang et al. 2023	Jiang et al. 2023
Area 3b	collapsed Area 1	collapsed Area 1	Jiang et al. 2023 (collapsed)
S2	same as Area 1	same as Area 1	adapted from Area 1
Thalamus 4	Jiang et al. 2023	Jiang et al. 2023 partly assumed by the authors	?

3 Results

Present the main findings, including figures and tables.

4 Discussion

Interpret the results and discuss their implications.

5 Conclusion

Summarize the main outcomes and future directions.

References